

Esophageal Cancer Treatment

Esophageal Cancer Treatment Categories

Category	AJCC	Stage	Treatment
High-grade Dysplasia	0	Tis	Radiofrequency Ablation
Superficial Tumors	I	T1a	Endoscopic Therapy
Localized Tumors	IIB	T1b T2	Surgery
Locally-advanced	III	T3 or N ⁺	Chemo ± RT → Surgery
Extra-Regional	IVA	N3	Chemo→ ChemoRT→ Surgery
Metastatic	IVB	M1	Chemotherapy ± Radiation

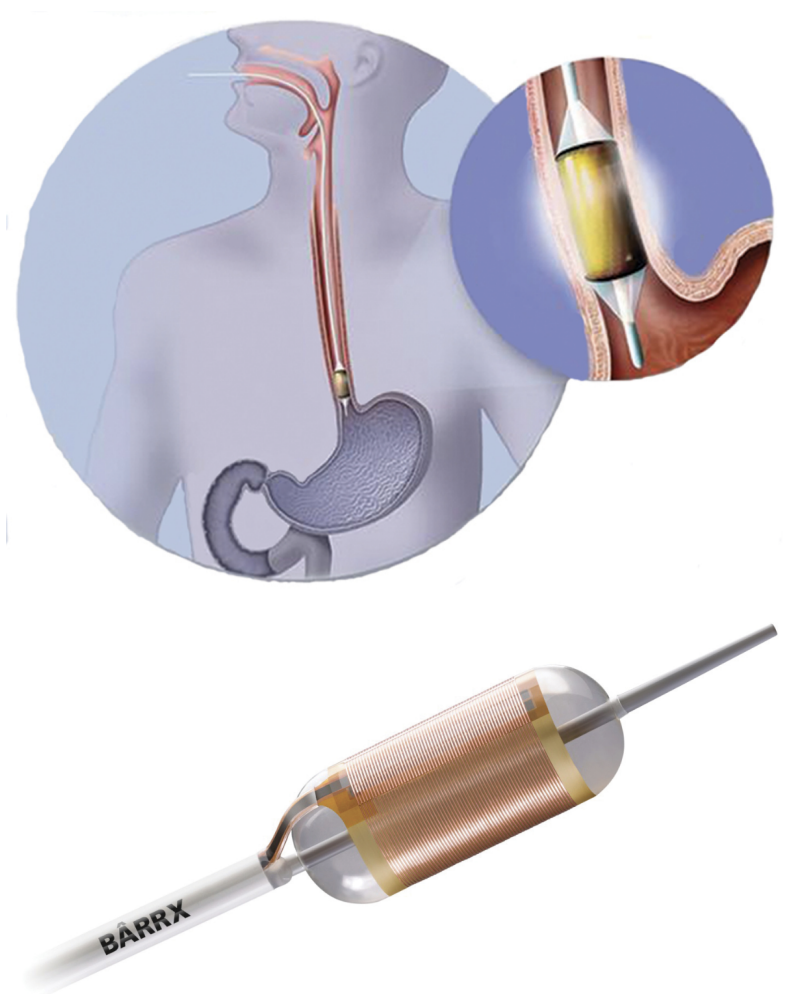
High-grade Dysplasia => Radiofrequency Ablation

127 patients with dysplasia randomized:

- Radio-frequency ablation
- Sham ablation

Low-grade dysplasia in 64

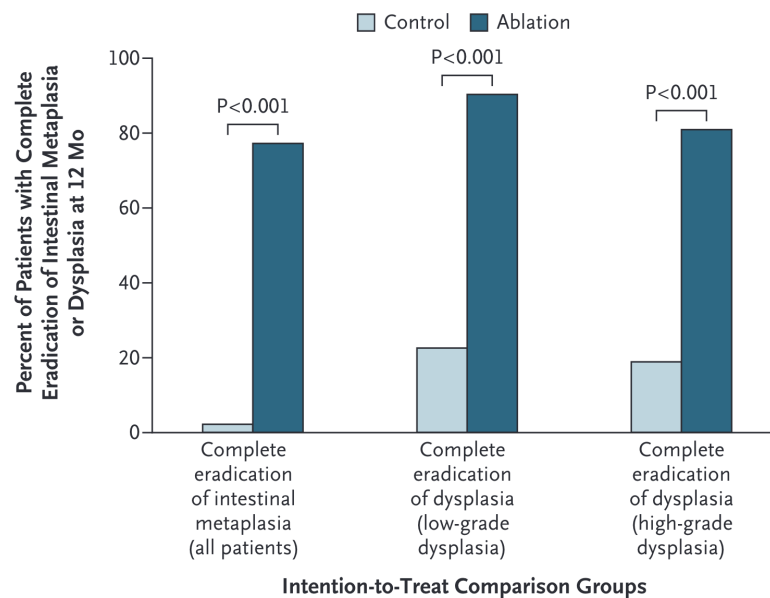
High-grade dysplasia in 63



(Shaheen et al. 2009)

Radiofrequency Ablation for Dysplasia

Radiofrequency Ablation results in eradication of Barrett's in 75% at 1 year



No role for primary surgery in high-grade dysplasia

(Shaheen et al. 2009)

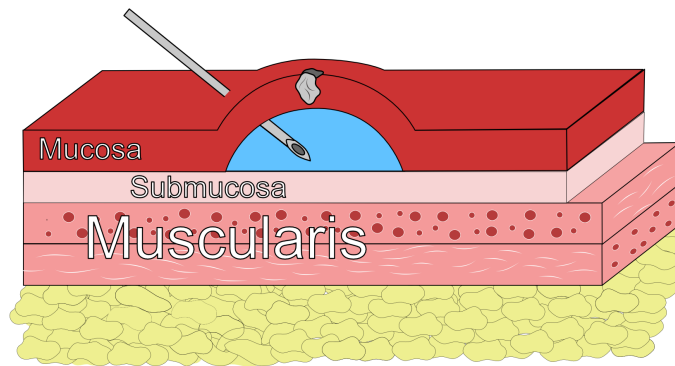
Superficial Tumors (T1)

Workup of nodular Barretts:

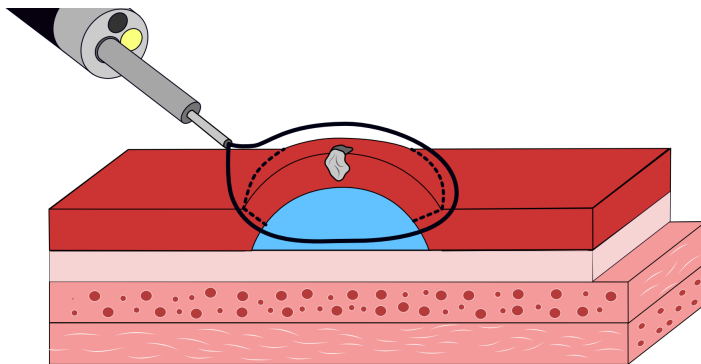
- Endoscopic Ultrasound
- Endoscopic Mucosal Resection
 - Diagnostic (T staging)
 - May be therapeutic for T1a tumors

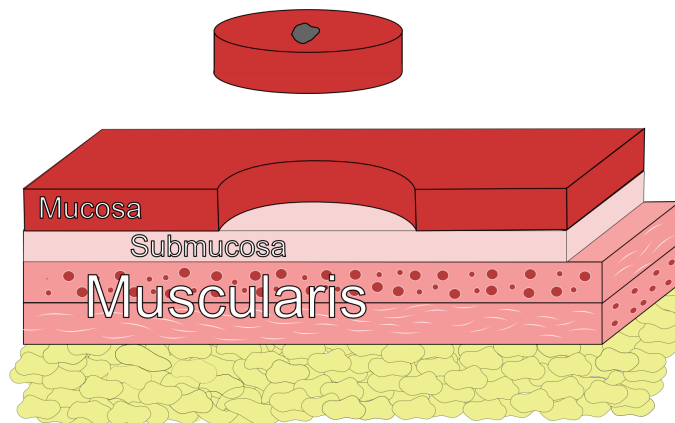
Endoscopic Resection for T1 Tumors

Tumor Lifted with submucosal injection



Endoscopic Mucosal Resection (EMR)



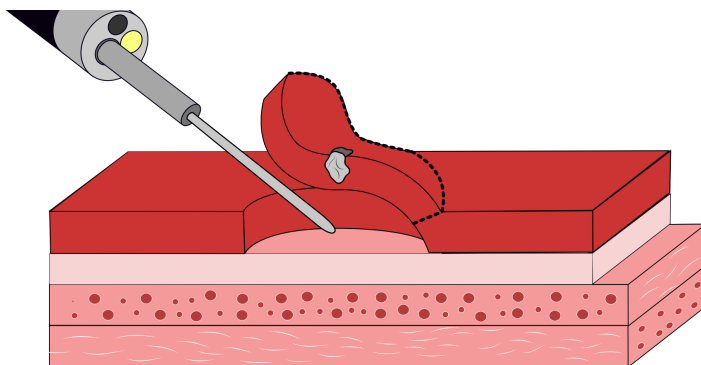


Endoscopic Submucosal Dissection for T1a/T1b

Endoscopic resection uses needle-knife to dissect *below* submucosa

Maybe suitable for T1b lesions *if* lesion is completely resected

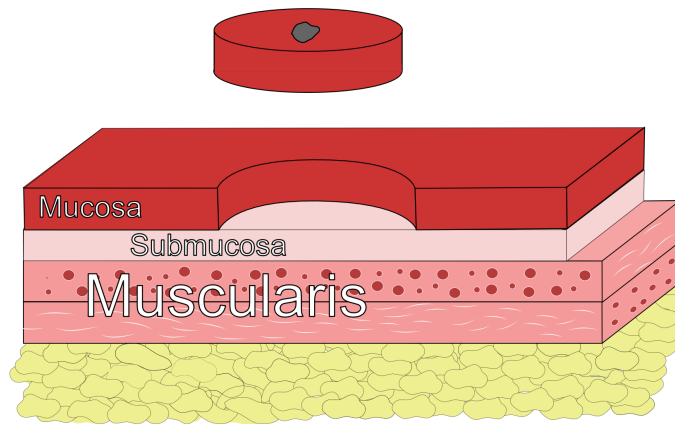
Risk of perforation



Favorable Features after Endoscopic Resection

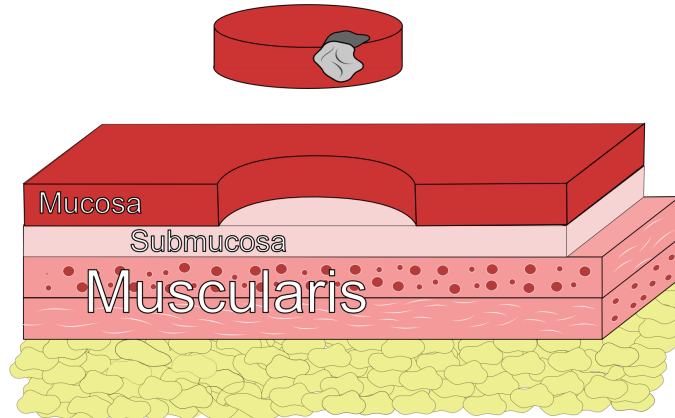
- Size <2cm
- No lymphovascular invasion
- Clear peripheral margin

- Clear deep margin



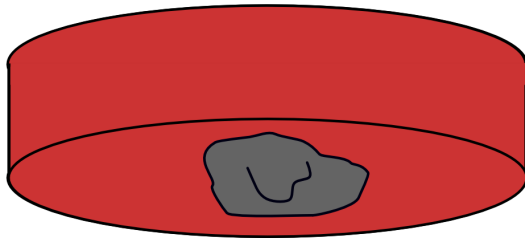
Unfavorable Features after Endoscopic Resection

- Positive lateral margin
- Lymphovascular invasion
- **Positive deep margin**



Positive Deep Margin after Endoscopic Resection

Associated with higher risk of recurrence AND lymph node metastasis



Localized Tumors (T2 N0)

Patients staged as uT2 N0 are candidates for primary surgery.
However:

- EUS has a 25% rate of understaging uT2 N0 tumors
- Understaged patients who undergo primary surgery would need chemo or chemoRT postop

Asymptomatic Tumors (minimal dysphagia)

- EUS to distinguish T2 from T3 tumors
- If uT2 N0 → CT chest/abdomen/pelvis → Esophagectomy
- If uT3 or N1 → PET → neoadjuvant therapy

Patients with dysphagia almost always are T3 tumors (and don't need EUS)

Patients with dysphagia can be *clinically* staged as T3 without EUS

Symptomatic Tumors (dysphagia)

Patients with dysphagia to solids or weight loss or tumor length >3cm are unlikely to have T1-2 tumors and can be initially evaluated with [PET Scan](#)

- Disease confined to the esophagus and regional nodes → [Locally-advanced](#)

- Metastatic disease → [Metastatic](#)
- N3 → [Extra-Regional](#)

EUS in Patients with Dysphagia

Memorial Sloan Kettering patients with esophageal cancer:

- 61 with dysphagia, 54 (89%) were found on EUS to have uT3-4 tumors.
- 53 without dysphagia, 25 (47%) were uT1-2 → candidates for primary surgery.

EUS can be omitted for patients with dysphagia

EUS can be useful in patients *without* dysphagia (to identify T2N0 → Surgery)

(Ripley et al. 2016)

PET Scan

PET has more specificity and sensitivity than CT in detecting regional lymph node and distal metastasis

Locally-advanced

Improved survival with adjunctive therapy. There are two options:

- ChemoRT → Surgery ([CROSS Trial](#))
- Chemo → Surgery → Chemo ([EsoPEC Trial](#))

CROSS Surgery vs ChemoRT → Surgery

368 esophageal cancer patients randomized:

- Surgery alone
- Chemo+RT → Surgery

75% adenocarcinoma. T3: 80% T2: 17%. age $\tilde{x}=60$

$\Rightarrow$ longer survival with Chemo+RT $\rightarrow$ Surgery

Chemotherapy: Weekly carboplatin and paclitaxel

Radiation: 4140 cGy in 23 fractions (180cGy/fraction)

(Shapiro et al. 2015)

CROSS Surgery vs ChemoRT $\rightarrow$ Surgery

	Surgery	CRT $\rightarrow$ Surgery	
Time to Surgery	24d	97d	
Unresectable	13%	4%	p=0.002
R0 Resection	92%	69%	p<0.001
Positive Nodes	75%	31%	
Complications	~	~	
Mortality	~	~	

CROSS - Overall Survival

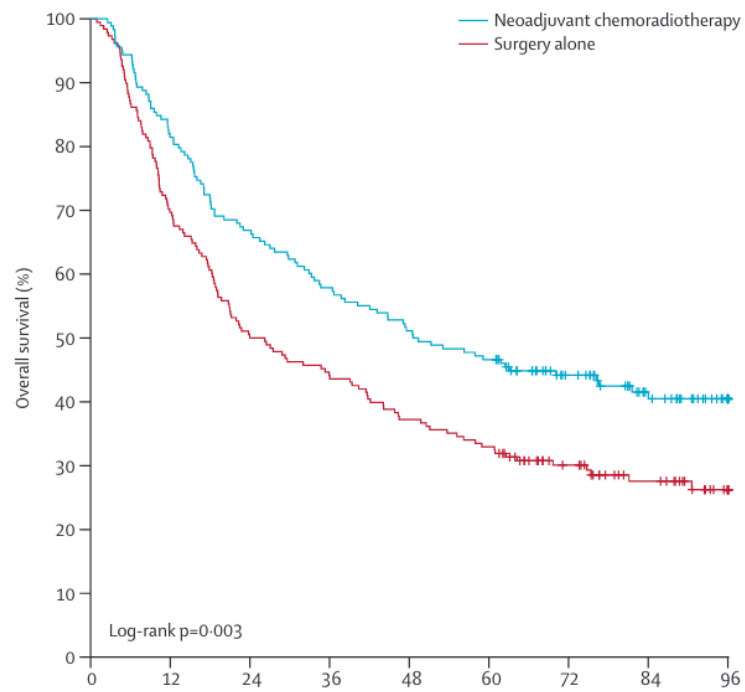


Figure 1: Surgery vs ChemoRT → Surgery

(Shapiro et al. 2015)

CROSS - Survival by Histology

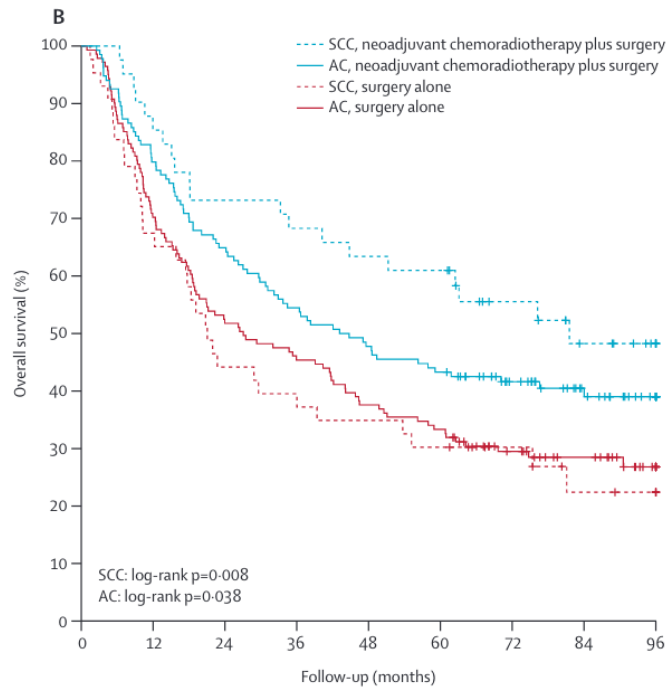


Figure 2: Surgery vs ChemoRT → Surgery

CROSS Effects by Histology

(Shapiro et al. 2015)

Adenocarcinoma

- Median survival 43mo vs 27mo
- Pathologic complete response in 23%

Squamous Cell Carcinoma

- Median survival 82mo vs 21mo
- Pathologic complete response in 49%

Surgical Therapy of Squamous Cell

(Shapiro et al. 2015)

Adjuvant Nivolumab after CROSS: Checkmate 577 Trial

EsoCA patients who received ChemoRT → Surgery with residual disease (not pCR)

Randomized to one year of nivolumab vs Observation

⇒ Adjuvant nivolumab group had longer median survival: 22mo vs 11mo

(Kelly et al. 2021)

Checkmate 577 Trial

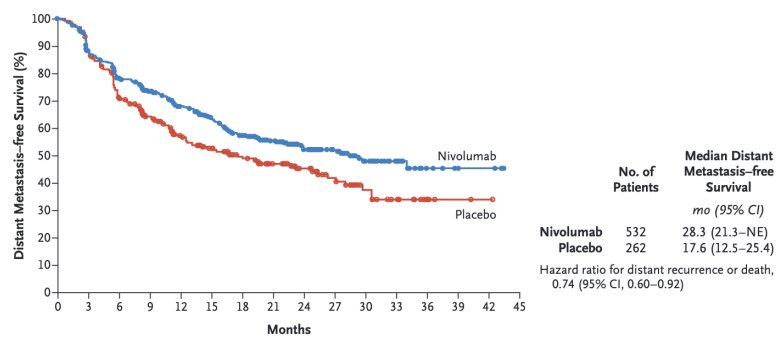


Figure 3: Adjuvant Nivolumab vs Observation

Nivolumab = Immune Checkpoint Inhibitor (ICI)

(Kelly et al. 2021)

PD-L1 agonist ligand

Interferes with tumor cell down-regulation of T cells

Active against stage IV esophageal cancer

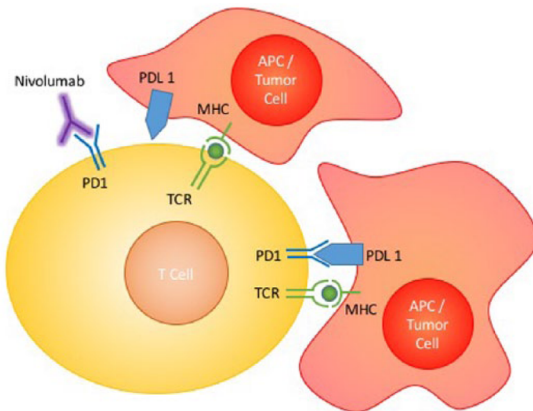


Figure 4: Nivolumab mechanism of action

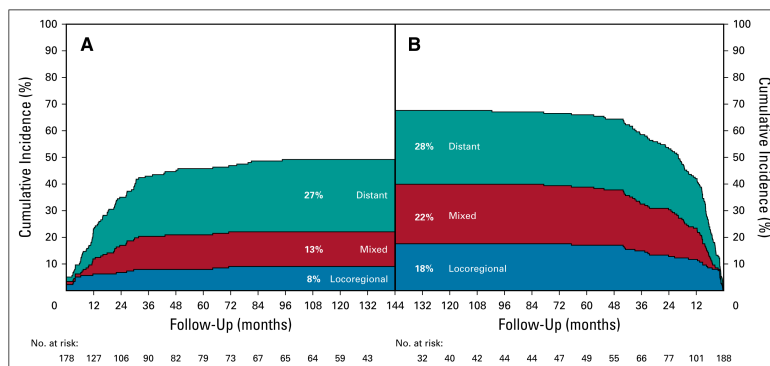
CROSS - ChemoRT Does Not Reduce Distant Failure

Sites of failure over time

ChemoRT + Surgery *vs* Surgery

ChemoRT appears to reduce risk of local or local+distant failure, but not isolated distant failure

⇒ Can chemotherapy reduce risk of distant failure?



(Shapiro et al. 2015)

EsoPEC CROSS vs FLOT for Adenocarcinoma

Adenocarcinoma esophagus and GE junction

- T1 N+ or T2-4a M0. Median age =63. 89% men

Randomized to CROSS (n=217) vs FLOT chemotherapy (n=221)

- CROSS: carboplatin/paclitaxel + 4140cGy → Surgery
- FLOT: FLOT → Surgery → FLOT

Excluded: Squamous cell, gastric cancer, T1N0, T4b, M1

(Hoeppner et al. 2025)

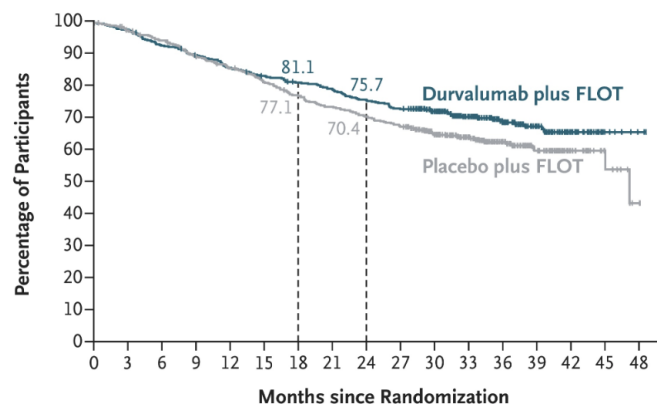
EsoPEC: FLOT superior to CROSS

	FLOT Periop Chemo	CROSS ChemoRT	CROSS Trial ChemoRT
90d Mortality	3.2%	5.6%	
$\tilde{x}$ Survival	66mo	37mo	43mo
3year Survival	57%	51%	58%
5year Survival	51%	29%	47%
path CR	17%	10%	23%

MATTERHORN Trial - FLOT ± Durvalumab

Durvalumab added to FLOT improved disease-free survival

A Overall Survival

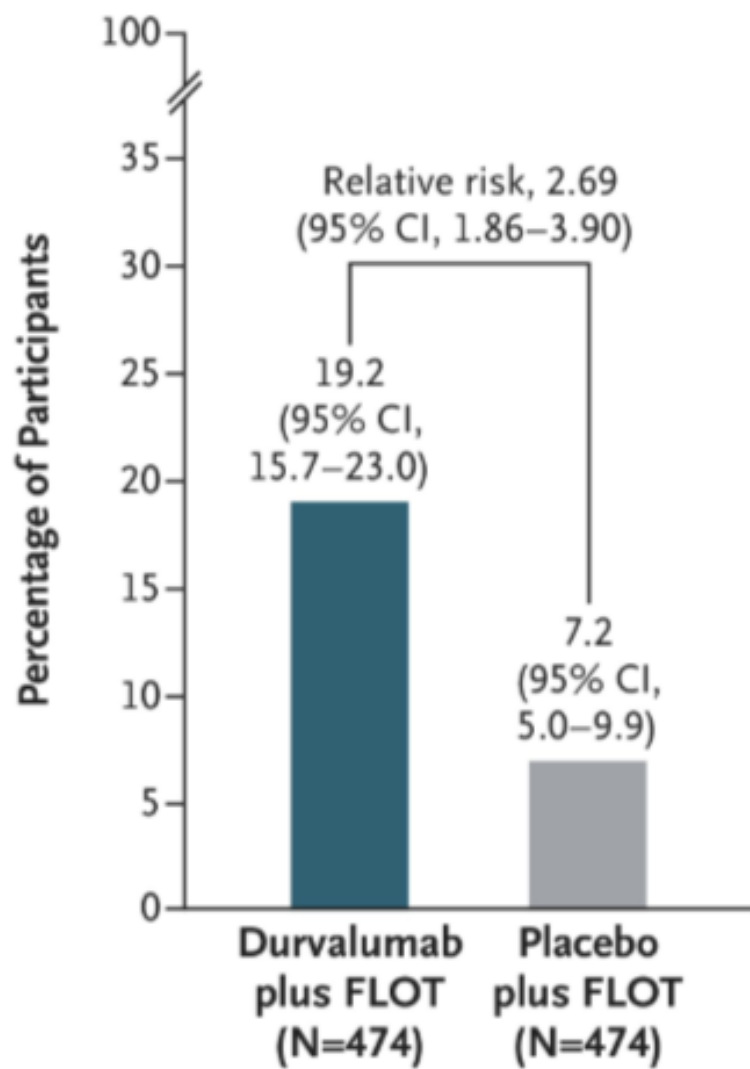


(Janjigian et al. 2025)

MATTERHORN Trial - FLOT ± Durvalumab

- pCR rate higher
- Survival benefit regardless of PD-L1
- Node-positive benefited more
- Diffuse-type histology did not benefit
- **Overall survival did not meet the pre-specified threshold**

B Pathological Complete Response



(Janjigian et al. 2025)

MSI-High Esophageal and GEJ Adenocarcinoma

MSI-Hi patients (15%) respond to immunotherapy

Neoadjuvant treatment with nivolumab + ipilumimab (3

months)

NEONIPIGA Clinical Trial

Surgery for Squamous Cell Carcinoma

Squamous Cell Carcinoma of the esophagus

- responds well to chemo+RT
- more difficult to get a surgical margin on the airway
- additional benefit of surgery on top of chemoRT is uncertain

Squamous Cell: ChemoRT ± Surgery (FFCD)

Squamous cell of the esophagus → 4500cGy RT + 2 cycles of cisplatin + 5FU

Patients with a clinical response were randomized:

- Surgery ⇒ 2 year survival 34% Median 17.7mo
- 3 cycles of chemo + 2000 cGy RT ⇒ 2 year survival 40% Median 19.3mo

⇒ No difference in overall survival

(Bedenne et al. 2007)

Squamous Cell: ChemoRT ± Surgery (German)

Squamous cell of the esophagus randomized:

- 4000 cGY RT + Chemo → Surgery. 64% 2-year PFS. Mortality 12.8%
- 6500cGy RT + Chemo: 41% 2-year PFS. Mortality 3.5%

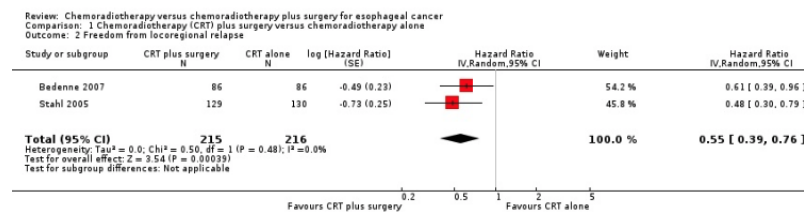
⇒ No difference in overall survival

(Stahl et al. 2005)

Freedom from Locoregional Relapse

Pooled data from Bedenne and Stahl

Surgery group has better freedom from locoregional relapse



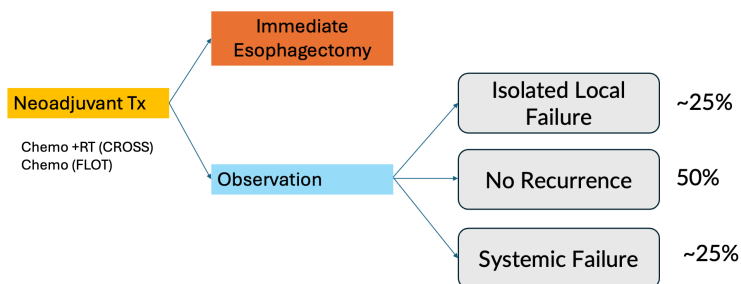
Non-Operative Management of Squamous Cell Carcinoma

CROSS Trial: ChemoRT → Surgery

Pathologic Complete Response = 49%

⇒ Is there a role for *non*-operative management of esophageal squamous cell carcinoma after chemoradiation?

Squamous Cell Carcinoma Treated with ChemoRT Alone



Surgery As Necessary Only (SANO) Clinical Trial

Gold Standard is chemoRT → surgery

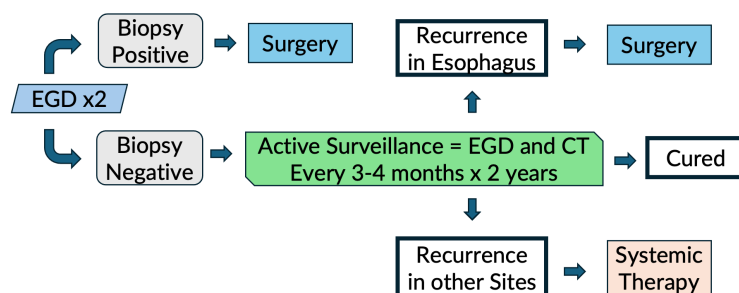
Control group was treated with immediate surgery after ChemoRT



Surgery As Necessary Only (SANO) Clinical Trial

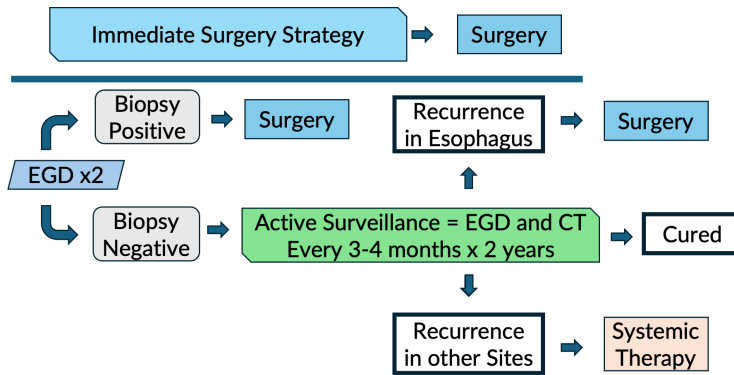
Experimental group: After ChemoRT, restaging with EGD x2 and PET

- Persistent disease → Surgery
- Complete Clinical Response (cCR): Negative EGD x2 and negative PET
 - Surveillance with periodic EGD and CT
 - Surgery only for an isolated local recurrence



(bondi-simpson275?)

Surgery As Necessary Only (SANO) Clinical Trial



Surgery As Necessary Only (SANO) Clinical Trial

Active surveillance strategy was non-inferior to immediate esophagectomy

- Surgery Group: 53 month median survival
- Active Surveillance Group: 43 month median survival

Sensitivity analysis shows histology did not alter conclusions

(bondi-simpson275?)

Role for Surgery after Preoperative Therapy

AATS Consensus guidelines: Esophagectomy is reasonable after in patients with a complete clinical response (cCR) after neoadjuvant therapy:

- Radiographic and endoscopic restaging is not adequately sensitive

Among SANO patients with cCR (negative EGD x2 and response on PET):

- 60% had residual disease

- 16% Node-positive

(molena884?)

Extra-regional Disease = N3

Patients with N3 disease are Stage IVA and have a poor prognosis.

Treatment consists of chemotherapy followed by radiation.

Patients with persistent disease in the esophagus (and no distant disease) would then be candidates for surgery

Metastatic = M1 IVB

FOLFOX is first-line systemic therapy for metastatic GI cancers

- Dose-limiting toxicity is frequently peripheral neuropathy

Adjuvant Immunotherapy: Checkmate 577 Trial

Immunotherapy with nivolumab as adjuvant therapy after CROSS regimen for patients with residual disease

Stage II/II Esophageal or GE junction cancers Adenocarcinoma or squamous cell

ChemoRT → Surgery *with residual disease on pathology*

Treatment Group: Nivolumab every 2 weeks x 4 months → every month x 8 months

vs

Control Group: No adjuvant therapy

⇒ Better survival in group with adjuvant nivolumab

(Kelly et al. 2021)

Neoadjuvant Chemo for EsoCA

- MAGIC trial (gastric): ECF¹→Surgery→ECF *vs* Surgery
- **OEO2 Trial**: (esophageal) Chemo→Surgery→ Chemo *vs* Surgery
- FLOT (gastric): FLOT²→Surgery→ FLOT *vs* ECF→Surgery→ECF
- **EsoPEC**: (esophageal):FLOT→Surgery→FLOT *vs* ChemoRT→Surgery (CROSS)

OEO2 Clinical Trial

- 802 Esophageal adenocarcinoma and squamous cell
- Randomized to Chemo → Surgery → Chemo *vs* Surgery alone
- Chemotherapy with ECF (Epirubicin, Cisplatin, 5FU)
- 5-year survival 23% for chemo+surgery vs 17% for surgery (HR 0.84 p=0.03)

(Allum et al. 2009)

Neo-Aegis Trial CROSS vs MAGIC/FLOT

- **Adenocarcinoma** T2-3 N0-3 M0 Tumor length <8cm
- ChemoRT arm: carboplatin + paclitaxel + 4140cGy
- Chemo arm: MAGIC (ECF) or FLOT (later in trial)
- No difference in overall survival
- R0 resection 96% with CROSS vs 82% with chemo
- pCR 12% with CROSS vs 4% with chemo

(reynolds1015?)

¹Epirubicin, Cisplatin, 5FU

²5FU, Leuovorin, Oxaliplatin, Decetaxol

Orientation Manual



References

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